Corporate Bond Yield & Liquidity Prediction using Machine Learning

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Objective

Why Predict Bond Yields & Liquidity?

- Pricing: Identify mispriced bonds
- Trading: Estimate transaction costs
- Risk: Portfolio VaR calculations
- Challenge: Non-linear relationships between credit risk, rates, and liquidity

CORPORATE BOND COMPLEXITY:

- Over-the-counter trading (illiquid)
- Embedded options (callable/convertible)
- Limited fundamental data (21% coverage)
- Non-linear price-yield dynamics

DATA & METHODOLOGY

4 Data Sources → Comprehensive Dataset:

1. WRDS TRACE: 78K bonds, monthly prices
2. Mergent FISD: Bond characteristics
3. Compustat: Issuer fundamentals (21%)
4. FRED: 21 macro variables

→ 2.3M observations, 2015-2024

FEATURE ENGINEERING: 226 features

- Momentum (ytm lags, returns)
- credit risk (ratings, changes)
- Liquidity (volume, spreads)
- Macro (rates, VIX, credit spreads)

MODEL: LightGBM Gradient Boosting

- Handles missing data natively
- Captures non-linear interactions
- 50 trials hyperparameter tuning

Results

MODEL PERFORMANCE: YIELD PREDICTION

Model	Test R^2	Test RMSE	Interpretation
Elastic Net	-0.40	184 bps	FAILED (worse than mean)
LightGBM Baseline	0.9125	46 bps	Strong performance
LightGBM TUNED	0.9227	43 bps	✓ BEST (92% variance explained)

LIQUIDITY PREDICTION (Bid-Ask Spreads)

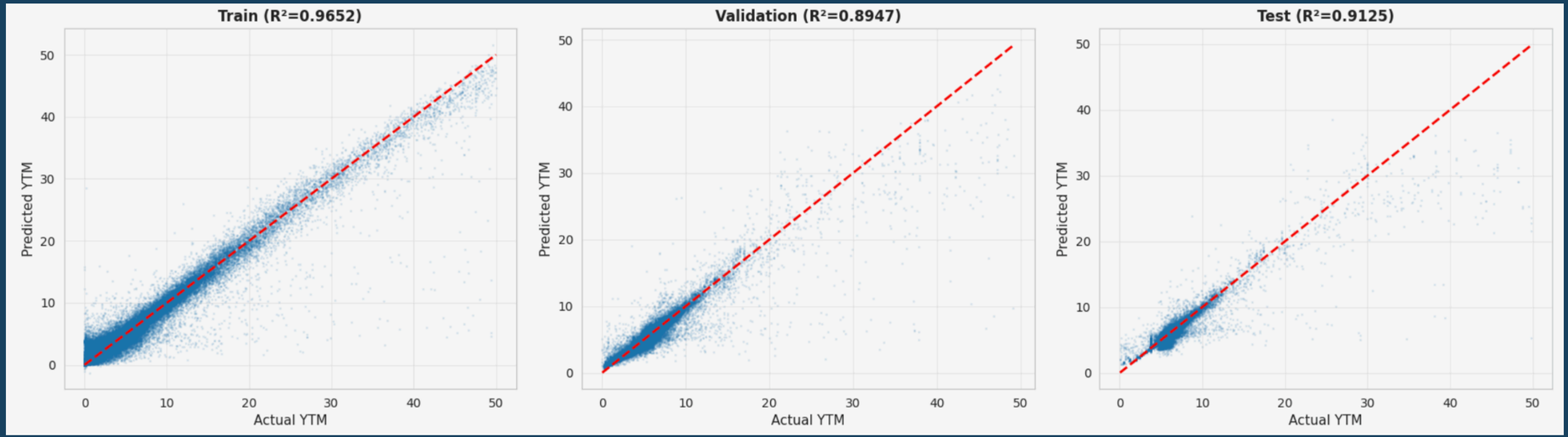
LightGBM Regression	0.081	1.21 bps	4.5× better than linear
Logistic (Binary)	70-77%	accuracy	Useful for liquid/illiquid

KEY TAKEAWAYS

YIELD: 92% R^2 | 43 bps error | Best for investment-grade (18 bps error)
Error increases with maturity: 0-3Y (7 bps) → 30Y+ (26 bps)

LIQUIDITY: 12.4% R^2 | 1.21 bps error | Better for IG (21% R^2) vs HY (10%)
Binary classification 70-77% accurate for liquid/illiquid screening

LINEAR MODELS FAIL: Elastic Net negative R^2 for yields, near-zero for liquidity



Insights

TECHNICAL RIGOR

Data Leakage Detection

- Initial R^2=0.99 triggered investigation
- Found 5 features containing target
- Proper validation with lagged features

Time-Based Validation

- Train: 2015-2021 (68%)
- Val: 2022-Jun 2023 (20%)
- Test: Jul 2023-Aug 2024 (12%)
- No look-ahead bias

Hyperparameter Optimization

- Bayesian search (Optuna)
- 50 trials, 90 minutes
- Optimal: 28 leaves, lr=0.06, L2=0.03

Smart Missing Data Handling

- 79% missing fundamentals
- LightGBM native handling
- Context-aware imputation

Impact & Limitations

YIELD MODEL (92% R^2):

- Relative Value Trading
- Flag bonds mispriced >50 bps
- 43 bps error provides buffer
- Risk Management
- VaR calculations (no parametrics)
- Portfolio stress testing
- Portfolio Optimization
- Expected return forecasting

LIQUIDITY MODEL (12.4% R^2):

- Transaction Cost Estimation
- Predicted spreads for execution
- Pre-Trade Screening
- 70-77% accuracy liquid/illiquid
- Liquidity Risk Management
- Identify illiquid exposures

LIQUIDITY PREDICTION RESULTS

- Performance: - LightGBM: 12.4% R^2 | 1.21 bps RMSE (7× better than linear baseline)
- Binary Classification: 70-77% accuracy (Useful for liquid/illiquid screening)
- By Segment:
  - Investment Grade: 21% R^2
  - High Yield: 10% R^2
- Long maturity more predictable

LIMITATIONS

- Distressed bonds (YTM >20%)
- Monthly frequency only
- Liquidity inherently noisy
- Requires retraining for regimes

Final takeaway: Gradient boosting achieves 92% R^2 for yields (state-of-the-art: equity models 1-3%, spread models 30-50%) and 12.4% R^2 for liquidity (7× better than linear baselines). Features are economically interpretable—yield persistence, credit quality, price dynamics. Non-linear methods essential: linear models failed completely. Rigorous validation critical: data leakage detection prevented overfitting.

Thank You

Q&A?